

Manuscript Tables

Stage 1: Scoping review and taxonomy

Table 1

Search Rounds and Database Yields

Round	Period	New records	Purpose
1	May to September 2021	1,040	Initial broad search; baseline coverage of the responsible gambling evidence base
2	October 2022 to January 2023	218	Refined search strings; capture of 2021 to 2022 literature
3	February to May 2024	333	Final update; capture of 2023 to 2024 literature
Unassigned	Not applicable	285	Round of entry could not be unambiguously confirmed against publication year
Total		1,876	

Note. All three databases (PsycINFO, MEDLINE, Web of Science Core Collection) were searched in every round. Round 3 included a Google Scholar sensitivity check. Round 3 search strings are available in Appendix 01.1.

Table 2*Taxonomy Domain Composition by Implementation Readiness*

Taxonomy domain	k	%	Dep	Val	Test	Conc
Messaging and Communications (MSG-*)	74	18.0	29	4	39	2
Therapeutic Interventions (THER-*)	57	13.9	5	45	6	1
Limit setting (LIMIT-*)	36	8.8	20	2	2	12
Compound (multi-tool records)	34	8.3	19	2	9	4
Structural Game Modifications (STRUCT-*)	33	8.0	10	6	10	7
Self-Regulation Strategies (SELF-REG)	30	7.3	0	5	0	25
Personalised Feedback (FEED-*)	29	7.1	16	7	2	4
Exclusion and Account Controls (EXCL-*)	29	7.1	25	0	0	4
Education and Information (EDU-*)	28	6.8	5	6	14	3
Proactive Contact (CONTACT-PROACTIVE)	13	3.2	5	8	0	0
Low-Risk Gambling Guidelines (GUIDE-LRGL)	13	3.2	0	0	0	13
Detection and Monitoring Systems (DETECT)	11	2.7	2	0	0	9
Mandatory breaks (BREAK-FORCE)	10	2.4	5	0	5	0
Multi-Component Programs and System Tools (PROGRAM)	7	1.7	4	0	0	3
Self-Assessment Tools (ASSESS-SELF)	6	1.5	3	0	0	3
Total	410	100.0	148	85	87	90

Note. Cells are tool-record counts. Implementation readiness records how far each tool was tested or deployed: Dep = Deployed, Val = Validated, Test = Tested, Conc = Conceptual (defined in Appendix 02.5). Percentages are of 410 tool records.

Table 3*Evidence Profile: Evidence Quality by Direction of Finding*

Quality level	Positive	Mixed	Negative	Null	Total
HIGH	0	0	0	0	0 (0.0%)
MODERATE-HIGH	11	1	1	1	14 (3.4%)
MODERATE	171	22	7	1	201 (49.0%)
LOW-MODERATE	147	29	7	2	185 (45.1%)
LOW	7	1	2	0	10 (2.4%)
Total	336 (82.0%)	53 (12.9%)	17 (4.1%)	4 (1.0%)	410 (100.0%)

Note. Cells are tool-record counts (N = 410). Evidence Quality levels (HIGH to LOW) are defined in Appendix 02.5 and Direction of Finding categories in Appendix 02.4. The HIGH tier is retained to preserve the scale ceiling (k = 0).

Stage 2: Framework consolidation

Table 4

Framework Domain Composition by Evidence Status

Framework domain	<i>k</i>	%	ET	TD	PS	ND
Financial Consumption Limits	23	11.0	14	4	5	0
Time and Session Management	19	9.0	13	5	1	0
Exclusion and Account Controls	15	7.1	11	1	3	0
Messaging and Communications	23	11.0	21	1	1	0
Personalised Feedback	9	4.3	9	0	0	0
Detection and Monitoring Systems	7	3.3	7	0	0	0
Self-Assessment Tools	6	2.9	6	0	0	0
Education and Information	13	6.2	11	0	2	0
Proactive Contact	5	2.4	5	0	0	0
Therapeutic Interventions	16	7.6	16	0	0	0
Self-Regulation Strategies	10	4.8	10	0	0	0
Structural Game Modifications	15	7.1	7	1	7	0
Low-Risk Gambling Guidelines	6	2.9	5	1	0	0
Multi-Component Programs and System Tools	7	3.3	6	0	1	0
Newly Designed Tools	36	17.1	2	0	0	34
Total	210	100.0	143	13	20	34

Note. Framework tools are grouped into 15 framework domains by primary mechanism. Evidence Status: ET = Empirically tested, TD = Theoretically derived, PS = Policy or structural, ND = Newly designed, recorded as Extrapolated in the deposited files (all defined in Appendix 04.2). Percentages are of 210 framework tools. The Newly Designed Tools domain comprises 34 Newly designed tools; the two others in that block (T200, T207) carry Empirically tested status and are counted as such.

Table 5
Framework Tool Specification Field Distributions

Delivery Context	Criterion	<i>k</i>	%
<i>Online; In-product</i>	Delivered within an online gambling platform while the gambler gambles or accesses their account.	117	55.7
<i>Cross-channel; In-product</i>	Operates across online and land-based contexts, typically via registry or cross-platform mechanisms.	56	26.7
<i>Land-based; In-product</i>	Delivered within a physical venue's gambling products or systems.	11	5.2
<i>Not channel-specific; Community</i>	Channel-agnostic; delivered through community-facing channels such as campaigns, materials, or helplines.	9	4.3
<i>Not channel-specific; Player-applied</i>	Channel-agnostic; the gambler implements the tool without operator or clinician infrastructure.	9	4.3
<i>Not channel-specific; Clinical</i>	Channel-agnostic; delivered through clinical or treatment contexts such as therapy or counselling.	5	2.4
Other (small categories)	Residual delivery contexts with fewer than five framework tools each.	3	1.4
Total		210	100.0

Governance	Criterion	<i>k</i>	%
<i>Voluntary (opt-in)</i>	Requires active gambler choice to enable.	95	45.2
<i>Mandatory</i>	Cannot be declined by the gambler.	82	39.0
<i>Voluntary (self-initiated)</i>	Applied independently by the gambler without operator infrastructure.	16	7.6
<i>Operator-side</i>	Operates entirely behind the scenes without gambler-facing controls.	9	4.3

Governance	Criterion	<i>k</i>	%
<i>Voluntary (opt-out)</i>	Enabled by default; requires gambler choice to disable.	4	1.9
<i>Hybrid</i>	Combines gambler-side choice with operator-side enforcement.	4	1.9
Total		210	100.0
Personalisation	Criterion	<i>k</i>	%
<i>Non-tailored</i>	Identical content or parameters for all gamblers who receive the tool.	66	31.4
<i>System-tailored</i>	Content or parameters tailored algorithmically from observed gambler data.	56	26.7
<i>User-tailored</i>	The gambler sets their own parameters or selects from options.	48	22.9
<i>User-tailored + System-tailored</i>	The gambler sets a goal or parameter; the system tailors delivery.	23	11.0
<i>Practitioner-tailored</i>	A clinician, advisor, or trained staff member adapts the tool by judgement-based assessment.	14	6.7
Other compound modes	Residual combinations of personalisation modes with fewer than three framework tools each.	3	1.4
Total		210	100.0
Trigger Stage	Criterion	<i>k</i>	%
<i>On-demand</i>	Available whenever the gambler seeks it, at their point of access.	67	31.9
<i>During-play</i>	Operates while the gambler is actively gambling within a session.	43	20.5
<i>Continuous</i>	Operates at all times, independent of session boundaries.	40	19.0
<i>Between-sessions</i>	Operates between gambling sessions, at account level rather than in play.	32	15.2
<i>Pre-session</i>	Operates before a gambling session begins.	22	10.5

Trigger Stage	Criterion	<i>k</i>	%
<i>Post-session</i>	Operates after a gambling session ends.	4	1.9
Other stages	Residual activation stages grouped for reporting.	2	1.0
Total		210	100.0

Note. Delivery Context categories with < 5 framework tools are grouped as ‘Other’. Personalisation modes with < 3 framework tools are grouped as ‘Other’. Full definitions and exemplar tools for every value are in Appendix 04.2.

Table 6
Evidence Status by Target Population

Target Population	ET	TD	PS	ND	Total
<i>Universal</i>	94	11	17	22	144 (68.6%)
<i>High-risk</i>	28	0	3	0	31 (14.8%)
<i>Moderate-High-risk</i>	14	0	0	8	22 (10.5%)
<i>Moderate-risk</i>	5	2	0	3	10 (4.8%)
<i>Young adults</i>	1	0	0	0	1 (0.5%)
<i>New accounts</i>	0	0	0	1	1 (0.5%)
<i>Research / system development</i>	1	0	0	0	1 (0.5%)
Total	143	13	20	34	210 (100.0%)

Note. Cells are framework tool counts. Evidence Status categories (ET, TD, PS, ND) are defined in Table 4. Only 5 of 143 Empirically tested framework tools target Moderate-risk populations specifically; three of the 10 Moderate-risk tools are Newly designed. This concentration on Universal targeting at the Empirically tested tier reflects the prevention-paradox pattern documented in Appendix 03.3. Counts record the population a tool was designed for, not the population it reaches. Universal tools are available to moderate-risk gamblers.

Stage 3: Implementation Architectures

Table 7

Tool Variant Distribution by Risk Tier

Architecture tool	UK	NL
<i>Deposit Limit</i>	10	10
<i>Loss Limit</i>	12	8
<i>Bet Limit</i>	8	8
<i>Login/Playtime Limit</i>	6	8
<i>Deposit Count Limit</i>	8	8
<i>Session Cooldown</i>	10	10
<i>Scheduled Break</i>	5	5
<i>Reality Check</i>	5	5
<i>RG Homepage</i>	5	5
<i>Cheat Sheet</i>	5	5
<i>Positive Play Self-Assessment</i>	5	5
<i>Dark Flow Self-Assessment</i>	5	5
<i>Bonus Ban</i>	5	0
<i>Promo Ban</i>	5	0
<i>Automatic Withdrawal Limit</i>	0	5
Total tool variants	94	87
By risk tier		
<i>R1 (low)</i>	30 (31.9%)	28 (32.2%)
<i>R2 (moderate)</i>	38 (40.4%)	38 (43.7%)
<i>R3 (high)</i>	20 (21.3%)	19 (21.8%)
<i>Young-adult overlay</i>	6 (6.4%)	2 (2.3%)

Note. Cells are deployed tool-variant counts; 0 = tool not deployed in that architecture (UK excludes Automatic Withdrawal Limit on technical grounds; NL excludes Bonus Ban and Promo Ban on regulatory grounds). Risk tiers: R1 (low), R2 (moderate), R3 (high); the young-adult overlay applies independently of risk tier (UK uses age-band sub-tiers for Loss Limit). Percentages are of each architecture's total tool variants (94 UK, 87 NL). Full specifications are in Appendix 05.1/05.2.